TEMPORAL HOSPITAL ETU BED DEMAND FORECASTING AND OPTIMIZATION

Lochana J.L.S

IT22555526

BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Sri Lanka

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Lochana J.L.S

IT22555526

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	**Student ID**	**Signature**
Lochana J.L.S	IT22555526	

…… ……

Signature of the supervisor Date

(Mr. Kaniska Yapa)

ABSTRACT

Efficient management of patient inflow and bed allocation in Emergency Treatment Units (ETUs) is a critical challenge in modern healthcare systems. Hospitals often face unpredictable patient demand, leading to overcrowding, resource shortages, and increased pressure on medical staff. Traditional approaches rely heavily on manual processes and reactive decision-making, which are insufficient for handling dynamic and time-sensitive situations.

This research proposes an intelligent temporal ETU hospital bed demand forecasting and optimization system that integrates machine learning with automated decision-making. The system utilizes an ensemble approach combining the Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM) models to predict patient inflow for upcoming shifts. These predictions are then processed through a rule-based triage algorithm that dynamically allocates patients across standard ward beds, surge capacity, and external facilities.

The system is implemented using a full-stack architecture, including a Python Flask backend, React.js frontend, and MongoDB database. A user-friendly dashboard provides real-time insights, including predicted demand, occupancy levels, and actionable bed allocation strategies.

Results demonstrate that the proposed system effectively reduces overcrowding, improves resource utilization, and supports proactive decision-making in ETU environments. By bridging the gap between forecasting and operational planning, this research contributes to the development of intelligent, data-driven healthcare management systems.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ETU	Emergency Treatment Unit
TFT	Temporal Fusion Transformer
LSTM	Long Short-Term Memory
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
UI	User Interface
DB	Database
JSON	JavaScript Object Notation
ICU	Intensive Care Unit

1. INTRODUCTION

1.1 General Introduction

Emergency Treatment Units (ETUs) play a critical role in modern healthcare systems, acting as the first point of contact for patients requiring urgent medical attention. These units are responsible for handling a wide range of emergency cases, including accidents, sudden illnesses, and life-threatening conditions. Due to their nature, ETUs must operate efficiently under high-pressure conditions, ensuring that patients receive timely and appropriate medical care.

However, one of the most significant challenges faced by ETUs worldwide is patient overcrowding and inefficient bed management. Hospitals often experience unpredictable fluctuations in patient inflow, especially during peak hours, adverse weather conditions, or disease outbreaks. When the number of incoming patients exceeds the available bed capacity, it leads to overcrowding, increased waiting times, delayed treatment, and additional stress on medical staff. These issues can significantly affect the quality of healthcare services and patient outcomes.

Traditionally, hospital bed management relies heavily on manual processes, where nurses and administrative staff update bed availability based on patient admissions and discharges. While this approach may function under normal conditions, it becomes inefficient during high-demand situations. Manual data entry is prone to delays, human errors, and inconsistencies, resulting in inaccurate representations of real-time hospital capacity. As a result, decision-making becomes reactive rather than proactive, making it difficult for hospitals to prepare for incoming patient surges.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), there is growing interest in applying predictive analytics to healthcare systems. Time-series forecasting models have shown significant potential in predicting patient inflow based on historical data and external factors such as weather conditions, holidays, and seasonal trends. These predictive capabilities enable healthcare systems to anticipate future demand rather than responding only after overcrowding occurs.

Despite these advancements, a major limitation in existing solutions is the lack of integration between prediction and real-world decision-making. Many systems focus solely on forecasting patient arrivals without providing actionable strategies to manage hospital resources effectively. In practical scenarios, hospitals require not only predictions but also automated mechanisms to optimize bed allocation and patient routing based on current capacity constraints.

To address these challenges, this research proposes an intelligent temporal ETU hospital bed demand forecasting and optimization system. The system combines advanced machine learning techniques with a practical triage decision-making framework to support real-time hospital operations. By predicting upcoming patient inflow and dynamically reallocating existing patients to appropriate wards, surge beds, or external facilities, the system aims to reduce overcrowding and improve resource utilization.

Ultimately, this research contributes towards transforming traditional hospital management from a reactive approach into a proactive and intelligent system. By integrating prediction with automated decision-making, the proposed solution enhances operational efficiency, reduces the burden on healthcare staff, and ensures better patient care in emergency treatment environments.

1.2 Background Literature

The increasing demand for efficient healthcare services has led to significant research in the application of Artificial Intelligence (AI) and Machine Learning (ML) within hospital management systems. In particular, Emergency Treatment Units (ETUs) have gained attention due to their critical role in handling time-sensitive medical cases. Overcrowding and inefficient resource allocation in ETUs have been widely identified as major challenges affecting patient care and operational efficiency.

Several studies have explored the use of machine learning techniques to predict patient arrivals in emergency departments. Time-series forecasting models such as Long Short-Term Memory (LSTM) networks have been widely used due to their ability to capture temporal dependencies in sequential data. LSTM models are particularly effective in identifying short-term fluctuations in patient inflow, enabling hospitals to anticipate sudden increases in demand. Research has

demonstrated that LSTM-based systems can significantly improve prediction accuracy when compared to traditional statistical methods.

More recently, advanced architectures such as the Temporal Fusion Transformer (TFT) have been introduced for multi-horizon time-series forecasting. The TFT model is capable of handling complex datasets with both static and dynamic variables, including environmental factors such as weather conditions, holidays, and seasonal trends. This makes it highly suitable for healthcare applications, where patient arrival patterns are influenced by multiple external variables. Studies have shown that TFT provides improved interpretability and long-term forecasting performance compared to conventional deep learning models.

In addition to predictive modeling, optimization techniques have been applied to address hospital resource allocation problems. Mixed-Integer Linear Programming (MILP) has been widely used in operations research for optimizing bed allocation, patient flow, and scheduling within hospitals. These models aim to maximize resource utilization while minimizing patient waiting times and overcrowding. However, many of these approaches rely heavily on predefined constraints and accurate real-time data inputs. In practical hospital environments, maintaining up-to-date data is challenging due to the reliance on manual data entry processes.

Furthermore, existing hospital management systems often separate prediction and decision-making processes. While predictive models can estimate patient inflow, they do not typically provide actionable strategies for handling capacity constraints. On the other hand, optimization models require precise input data and may fail when dealing with highly dynamic and uncertain real-world conditions, such as sudden patient surges or incomplete data records.

Another important aspect highlighted in previous research is the dependency on manual data entry for tracking hospital capacity. Nurses and administrative staff are responsible for updating bed availability, admissions, and discharges. During high-pressure situations, this manual process becomes unreliable, leading to delays and inconsistencies in data. As a result, decision-making systems may operate on outdated or incorrect information, reducing their effectiveness.

Although significant progress has been made in both predictive analytics and hospital optimization, there is still a lack of integrated systems that combine these capabilities into a single,

autonomous solution. Existing approaches often address only one part of the problem, either focusing on prediction or optimization, without considering the practical workflow of hospital operations.

Therefore, there is a clear need for a unified system that can accurately forecast patient inflow while simultaneously providing real-time, practical solutions for bed allocation and patient routing. Such a system should be capable of handling uncertainty, reducing dependency on manual inputs, and supporting healthcare staff in making faster and more reliable decisions in emergency environments.

1.3 Research Gap

Despite the significant advancements in healthcare analytics and hospital resource management, several critical limitations remain in existing systems, particularly in the context of Emergency Treatment Units (ETUs). These limitations highlight the need for a more integrated and practical solution that bridges the gap between predictive modeling and real-world hospital operations.

Firstly, many existing studies focus primarily on predicting patient arrivals using machine learning techniques such as LSTM and other time-series models. While these approaches provide valuable insights into future patient inflow, they do not offer actionable solutions for managing hospital capacity. In real-world scenarios, hospital administrators require not only predictions but also clear and immediate strategies to handle incoming patients when bed capacity is limited.

Secondly, current hospital optimization systems often rely heavily on manual data entry to determine real-time bed availability. Nurses and administrative staff are responsible for updating patient admissions, discharges, and ward occupancy. During high-demand situations, this manual process becomes unreliable due to workload pressure, leading to delayed or inaccurate data. As a result, optimization models may operate on outdated information, reducing their effectiveness in critical decision-making.

Furthermore, most existing optimization approaches, such as Mixed-Integer Linear Programming (MILP), depend on strict mathematical constraints and external solvers. These methods can become unstable or infeasible when dealing with highly dynamic environments, such as sudden

patient surges or incomplete datasets. In emergency situations, where decisions must be made instantly, reliance on complex external solvers can lead to system failures or delays.

Another key limitation is the lack of demographic-specific routing in existing systems. Many predictive models generate aggregate patient counts without considering practical constraints such as gender-based ward allocation. In real hospital environments, patients must be assigned to appropriate wards based on such constraints, making it essential for any optimization system to incorporate these factors into its decision-making process.

Finally, there is a lack of fully autonomous systems that integrate prediction, capacity estimation, and decision-making into a unified framework. Existing solutions often treat these components separately, requiring human intervention to connect prediction outputs with operational decisions. This separation limits the ability of hospitals to respond proactively to changing conditions.

Therefore, there is a clear research gap in developing a unified, autonomous ETU management system that combines accurate patient inflow prediction with real-time, practical bed allocation and triage decision-making. Such a system should minimize reliance on manual data entry, handle uncertainty effectively, and provide actionable solutions that align with real hospital workflows.

1.4 Research Problem

Emergency Treatment Units (ETUs) operate in highly dynamic and unpredictable environments where patient inflow can fluctuate significantly due to factors such as time of day, weather conditions, and disease outbreaks. One of the most critical challenges faced by hospitals is the efficient management of limited bed capacity under these uncertain conditions. When the number of incoming patients exceeds available resources, it leads to overcrowding, delayed treatment, and increased pressure on healthcare staff.

Currently, hospital bed management is largely dependent on manual processes. Nurses and administrative personnel are responsible for recording patient admissions, discharges, and ward occupancy. However, during peak periods, these manual updates are often delayed or incomplete due to workload pressure. As a result, the available data used for decision-making may not accurately reflect the real-time status of hospital capacity. This leads to inefficient bed allocation

and reactive decision-making, where hospitals respond to overcrowding only after it has already occurred.

Although machine learning models have been introduced to predict patient inflow, these systems typically operate in isolation and do not provide practical solutions for managing capacity constraints. Predicting the number of incoming patients alone is insufficient, as hospitals still lack automated mechanisms to decide how to allocate beds or manage patient overflow effectively.

Furthermore, existing optimization techniques often rely on rigid mathematical models and external solvers, which may fail under real-world conditions involving incomplete data or extreme patient surges. These systems also do not adequately consider practical constraints such as gender-based ward allocation and the availability of surge beds, which are essential in real hospital environments.

Therefore, the core research problem addressed in this study is:

How can a hospital autonomously predict incoming patient demand and dynamically optimize ETU bed allocation in real time, while minimizing reliance on manual data entry and ensuring practical, constraint-aware patient routing?

To address this problem, it is necessary to develop an integrated system that combines predictive modeling, real-time capacity estimation, and automated triage decision-making. Such a system should be capable of handling uncertainty, adapting to dynamic hospital conditions, and providing actionable solutions that support healthcare staff in maintaining efficient patient flow within Emergency Treatment Units.

1.5 Research Objectives

The primary objective of this research is to design, develop, and evaluate a temporal ETU hospital bed demand forecasting and optimization system that can autonomously support real-time decision-making in Emergency Treatment Units. The system aims to enhance hospital efficiency by predicting patient inflow and dynamically optimizing bed allocation based on current capacity constraints.

To achieve this main objective, the following specific objectives have been identified:

1.5.1 Main Objective

- To develop an intelligent and autonomous system that predicts patient inflow and optimizes bed allocation in Emergency Treatment Units using machine learning techniques.

1.5.2 Specific Objectives

1. To develop a time-series forecasting model capable of predicting incoming patient demand for the next hospital shift using historical ETU data and contextual factors such as weather conditions, holidays, and shift patterns.
2. To implement an ensemble machine learning approach by combining Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM) models in order to improve prediction accuracy and stability.
3. To design a triage decision-making mechanism that calculates ETU overflow and determines how many patients need to be transferred based on real-time capacity constraints.
4. To develop a practical patient routing strategy that allocates patients to appropriate destinations, prioritizing standard ward beds, followed by surge beds, and finally external hospital transfers when necessary.
5. To incorporate real-world operational constraints, such as gender-based ward allocation, into the optimization process to ensure that the system aligns with actual hospital workflows.
6. To reduce dependency on manual data entry by enabling the system to utilize available hospital data sources for real-time decision support.
7. To implement a user-friendly interface that provides clear and actionable insights to healthcare staff, enabling proactive decision-making before patient overcrowding occurs.

2. METHODOLOGY

2.1 Introduction

This chapter presents the methodology adopted to design and implement the temporal ETU hospital bed demand forecasting and optimization system. The proposed system aims to address the limitations of traditional hospital management by integrating predictive analytics with automated decision-making to support real-time operations in Emergency Treatment Units (ETUs).

The methodology focuses on combining data-driven forecasting techniques with a practical triage and bed allocation mechanism. Unlike conventional approaches that rely solely on prediction or optimization, the proposed system integrates both components into a unified framework. This enables the system to not only anticipate incoming patient demand but also provide actionable solutions to manage hospital capacity effectively.

The overall system architecture is designed as a multi-layered structure consisting of a frontend interface, a backend processing unit, and a centralized database. The backend serves as the core component, where data is processed, predictions are generated, and optimization decisions are made. Machine learning models are used to forecast patient inflow, while a rule-based triage algorithm is implemented to handle bed allocation and patient routing.

The methodology also incorporates real-world hospital constraints, such as limited bed capacity, gender-based ward allocation, and the availability of surge beds. These constraints are essential to ensure that the system operates in a manner consistent with actual hospital workflows. Furthermore, the system is designed to minimize reliance on manual data entry by utilizing existing data sources and automated calculations wherever possible.

To achieve the objectives of this research, the methodology is divided into several key components, including system overview, data collection, data preprocessing, model design, triage logic, system implementation, and testing. Each of these components plays a crucial role in ensuring the accuracy, reliability, and practicality of the proposed solution.

Overall, this methodology provides a comprehensive framework for developing an intelligent and autonomous ETU management system that can improve hospital efficiency, reduce overcrowding, and support proactive decision-making in emergency healthcare environments.

2.2 System Overview

The proposed temporal ETU hospital bed demand forecasting and optimization system is designed as an intelligent, data-driven framework that integrates prediction, capacity analysis, and automated triage decision-making into a unified solution. The system transforms traditional reactive hospital management into a proactive approach by forecasting patient demand and preparing resources in advance.

The overall workflow of the system consists of three main stages: data acquisition, patient inflow prediction, and bed allocation with triage decision-making. These stages are interconnected and operate in real time to support efficient hospital operations.

2.2.1 System Workflow

The system begins by collecting relevant data required for prediction and decision-making. This includes historical patient admission records, current ETU occupancy levels, available bed capacity, and contextual factors such as weather conditions and shift timing. These inputs provide a comprehensive representation of the hospital's operational state.

Once the data is gathered, it is processed and fed into the machine learning models. The system utilizes an ensemble approach that combines Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM) models to forecast the number of incoming patients for the next shift. The prediction is generated based on both long-term patterns and short-term variations in patient inflow.

After predicting the expected number of incoming patients, the system evaluates the current ETU capacity. It calculates whether the predicted patient load can be accommodated within the available bed capacity. If the demand exceeds capacity, the system identifies the number of patients that cannot be accommodated, referred to as overflow.

The next stage involves the triage decision-making process. Instead of rejecting incoming patients, the system frees up space by transferring existing ETU patients to appropriate locations. The routing process follows a prioritized strategy:

- Patients are first transferred to available beds in standard hospital wards.
- If ward capacity is insufficient, surge beds are utilized as temporary accommodations.
- If both ward and surge capacities are exhausted, patients are recommended for transfer to external healthcare facilities.

Additionally, the system ensures that practical constraints are maintained during routing. For example, patients are allocated to wards based on gender-specific requirements, reflecting real hospital policies and operational conditions.

Finally, the system presents the results through a user interface, providing healthcare staff with a clear and actionable plan. This includes predicted patient inflow, current capacity status, and recommended patient transfers. By providing this information in advance, the system enables proactive decision-making and reduces the risk of overcrowding.

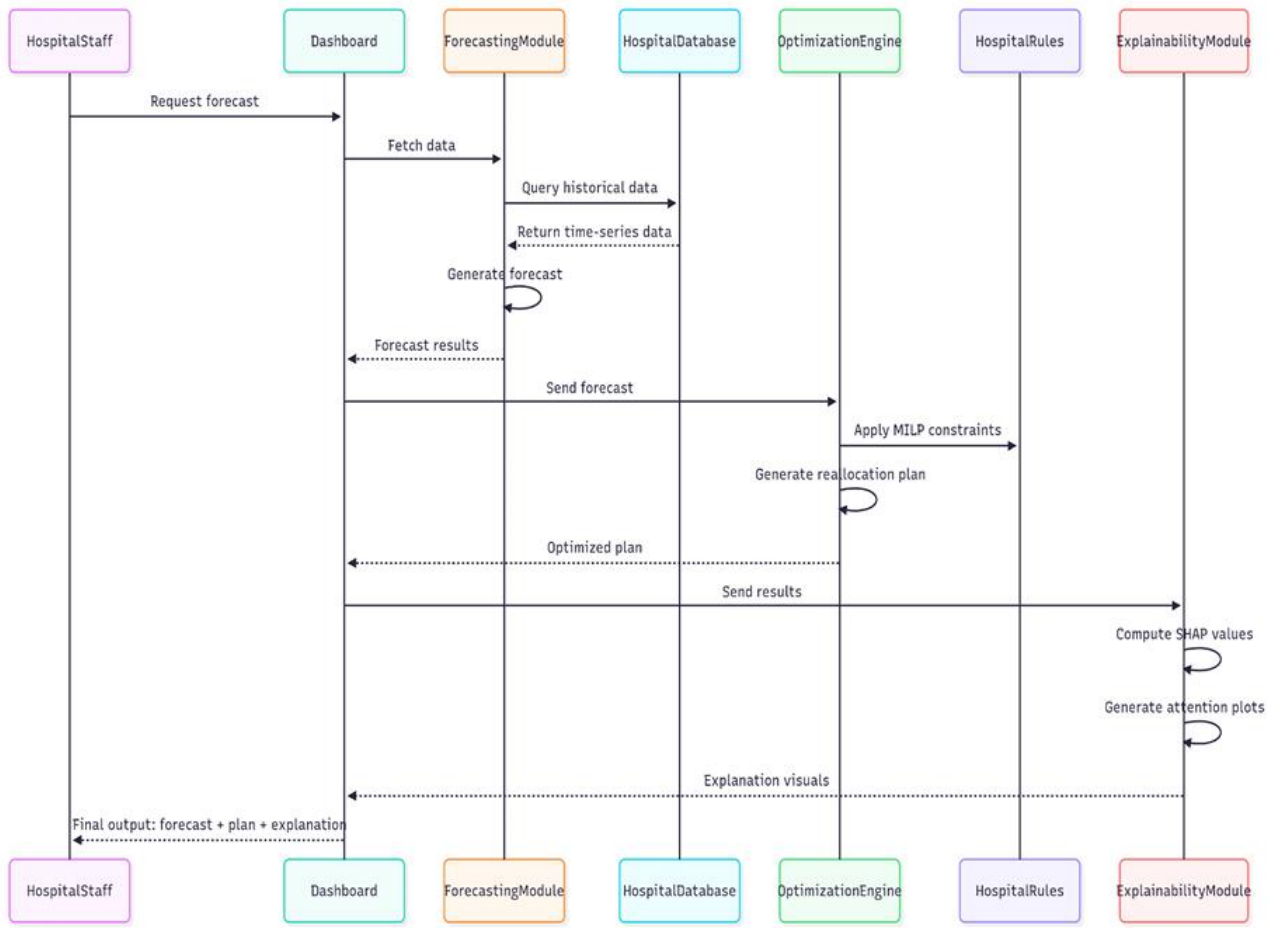

Figure 1- System workflow and interaction process

The workflow diagram illustrates the interaction between hospital staff, the dashboard, forecasting module, optimization engine, and explainability components. It shows how user requests trigger data retrieval, prediction generation, optimization, and final output delivery.

2.2.2 System Architecture

The system is implemented using a three-tier architecture consisting of frontend, backend, and database layers.

- **Frontend Layer:** The frontend is developed using React.js and serves as the user interface for healthcare staff. It displays real-time information such as predicted patient inflow, bed

availability, and recommended actions. The interface is designed to be simple and intuitive, allowing users to quickly understand the system's outputs.

- **Backend Layer:** The backend is implemented using a Python Flask framework. It acts as the core processing unit of the system, handling data retrieval, executing machine learning models, and performing triage decision-making. The backend also manages communication between the frontend and the database.
- **Database Layer:** The system uses MongoDB as a NoSQL database to store and manage hospital data. This includes historical patient records, daily ETU inputs, bed inventory information, and surge capacity details. MongoDB enables flexible and scalable data storage, allowing the system to handle dynamic and real-time data efficiently.

The interaction between these layers ensures seamless data flow and real-time processing. When the frontend requests data, the backend retrieves the latest information from the database, performs predictions and optimization, and returns the results to be displayed to the user.

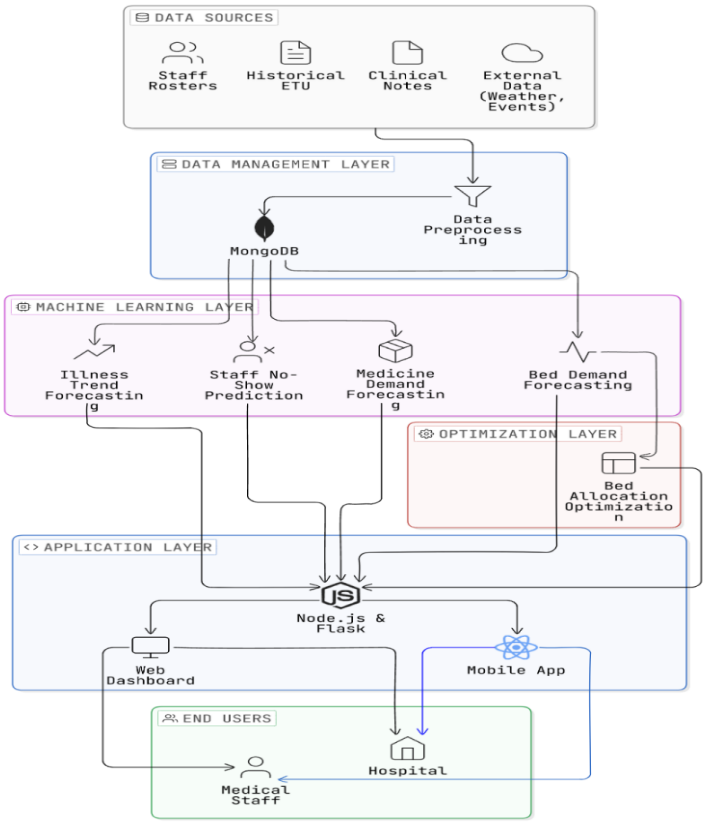

Figure 2- Proposed system architecture

The system architecture illustrates the layered design of the proposed ETU management system. It includes data sources, data management, machine learning forecasting modules, optimization logic, and application interfaces. The architecture demonstrates how raw hospital data is transformed into actionable insights through prediction and triage mechanisms.

2.3 Data Collection

The performance and reliability of the proposed system largely depend on the quality and structure of the dataset used for training and evaluation. Therefore, a comprehensive and well-structured dataset was utilized to capture real-world patterns of patient inflow in Emergency Treatment Units (ETUs).

The dataset used in this research consists of historical ETU records spanning a period from 2016 to 2025. It contains shift-level data, which reflects the operational reality of hospital workflows. Each day is divided into multiple shifts, such as morning, evening, and night, allowing the system to model patient inflow patterns more accurately based on time-specific variations.

A key characteristic of the dataset is its demographic segmentation. Patient data is recorded separately for male and female categories, which enables the system to generate gender-specific predictions. This feature is particularly important for practical hospital implementation, as bed allocation often depends on gender-based ward structures.

The dataset includes several important attributes that contribute to the prediction and decision-making processes. These attributes can be broadly categorized into operational data and contextual data.

Figure 3- Yaggahapitiya Divisional Hospital (data collection location)

Data for this research was collected based on observations and insights from Yaggahapitiya Divisional Hospital. This real-world healthcare environment provided practical understanding of ETU operations, patient flow, and bed management challenges, which were incorporated into the system design. This real-world exposure helped validate the assumptions used in the dataset and ensured that the proposed system reflects practical hospital workflows

2.3.1 Operational Data

Operational data represents the internal state of the hospital and includes:

- **ETU Admissions:** Number of patients arriving at the ETU during a specific shift.
- **ETU Discharges:** Number of patients discharged from the ETU.
- **ETU Occupied Beds:** Number of beds currently occupied in the ETU.
- **ETU Bed Capacity:** Total number of beds available in the ETU.

These attributes provide a direct representation of hospital capacity and patient flow dynamics.

2.3.2 Contextual Data

Contextual data captures external factors that may influence patient inflow patterns. These include:

- **Date and Day of the Week:** Used to identify weekly trends and recurring patterns.

- **Shift Information:** Indicates whether the data corresponds to morning, evening, or night shifts.
- **Weather Conditions:** Environmental factors such as rainy or sunny conditions, which may affect accident rates and patient arrivals.
- **Public Holidays:** Days that may experience variations in hospital activity.
- **Outbreak Alerts:** Indicators of disease outbreaks or special health conditions that may increase patient inflow.
- **Public Transport Status:** Conditions affecting mobility, which can influence the number of patients arriving at the hospital.

These contextual features enable the system to capture complex relationships between external factors and patient demand.

2.3.3 Data Characteristics

The dataset is structured as a time-series dataset, where each record represents a specific shift and includes both operational and contextual attributes. The total number of records exceeds several thousand entries, providing sufficient data for training machine learning models.

Additionally, the dataset reflects real-world hospital variability, including fluctuations in patient arrivals, differences between shifts, and irregular patterns caused by external events. This diversity ensures that the models trained on this dataset can generalize well to real operational scenarios.

2.3.4 Data Usage in the System

The collected dataset serves multiple purposes within the system:

- It is used to train the machine learning models (TFT and LSTM) for predicting future patient inflow.
- It provides historical trends that help the system understand seasonal and temporal variations.
- It supports the evaluation of model performance by comparing predicted values with actual patient arrivals.

By utilizing both operational and contextual data, the system is able to generate accurate and meaningful predictions, which form the foundation for subsequent bed optimization and triage decision-making processes.

2.4 Data Preprocessing

Data preprocessing is a critical step in the development of machine learning models, as the quality of input data directly affects the accuracy and reliability of predictions. The dataset used in this research contains both operational and contextual attributes, which require careful preparation before being used for model training and evaluation.

The preprocessing phase involves cleaning the data, handling missing values, transforming categorical variables, and preparing the dataset in a format suitable for time-series forecasting models such as Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM).

2.4.1 Data Cleaning

The raw dataset may contain inconsistencies such as missing values, incorrect entries, or incomplete records. These issues can negatively impact model performance if not addressed properly. Therefore, the dataset was first examined to identify and remove invalid or duplicate records.

In cases where numerical values such as admissions or discharges were missing, appropriate strategies were applied to ensure continuity of the data. Records with critical missing information were either corrected using available related fields or excluded if they could not be reliably reconstructed.

2.4.2 Handling Missing Values

Missing values were handled based on the type of data:

- **Numerical Fields:** Missing values in fields such as admissions, discharges, and occupied beds were filled using logical assumptions or derived values based on related columns. For

example, total occupancy could be calculated using gender-specific counts when necessary.

- **Categorical Fields:** Missing categorical values, such as weather or outbreak alerts, were replaced with a default category (e.g., "Unknown" or "None") to maintain consistency.

This approach ensures that the dataset remains complete without introducing significant bias.

2.4.3 Feature Transformation

Several features in the dataset required transformation to be suitable for machine learning models:

- **Date Conversion:** The date field was converted into a structured format to extract useful components such as day of the week and time sequence.
- **Shift Encoding:** Shift identifiers (morning, evening, night) were encoded into numerical or categorical representations to be used as model inputs.
- **Categorical Encoding:** Variables such as weather conditions, transport status, and outbreak alerts were encoded into numerical formats using techniques such as label encoding or one-hot encoding.

These transformations allow the models to interpret non-numeric data effectively.

2.4.4 Time-Series Structuring

Since the dataset represents time-dependent data, it was structured as a time-series sequence. Each record was ordered chronologically based on date and shift, ensuring that temporal relationships between observations were preserved.

For the LSTM model, sliding window techniques were applied to create input sequences. A fixed number of previous time steps (e.g., several past shifts) were used as input features to predict future patient inflow. This allows the model to learn patterns over time.

For the TFT model, both static and time-varying features were prepared to enable multi-horizon forecasting. This includes combining historical observations with known future inputs such as shift schedules and calendar-based features.

2.4.5 Normalization and Scaling

To improve model performance and convergence, numerical features were normalized or scaled. This process ensures that all input variables fall within a similar range, preventing certain features from dominating others during training.

Scaling techniques such as Min-Max normalization or standardization were applied to key numerical attributes, including admissions, discharges, and occupied beds.

2.4.6 Data Preparation for Model Training

After completing all preprocessing steps, the dataset was divided into training and testing subsets. The training set was used to train the machine learning models, while the testing set was used to evaluate their performance.

The prepared dataset was then formatted according to the requirements of each model:

- **LSTM Model:** Input sequences were structured as time-series arrays with multiple features per time step.
- **TFT Model:** Data was organized into a format suitable for multi-horizon forecasting, including both historical inputs and known future variables.

2.5 Model Design

The proposed system utilizes an ensemble machine learning approach to accurately forecast patient inflow in Emergency Treatment Units (ETUs). The model design combines the strengths of two advanced time-series forecasting techniques: the Temporal Fusion Transformer (TFT) and the Long Short-Term Memory (LSTM) network. This hybrid approach enables the system to capture both long-term patterns and short-term variations in patient arrival data.

The ensemble model is designed to improve prediction accuracy and stability by integrating outputs from both models. Each model contributes uniquely to the prediction process, addressing different aspects of temporal data behavior.

2.5.1 Temporal Fusion Transformer (TFT)

The Temporal Fusion Transformer (TFT) is a state-of-the-art deep learning model designed for multi-horizon time-series forecasting. It is particularly suitable for complex datasets that include both static and time-varying features.

In this research, the TFT model is used as the primary forecasting component due to its ability to capture long-term temporal dependencies and interpret complex relationships between variables. The model processes historical patient data along with contextual features such as weather conditions, day of the week, and shift information.

Key characteristics of the TFT model include:

- **Multi-Horizon Forecasting:** The ability to predict future values across multiple time steps.
- **Handling of Heterogeneous Data:** Integration of both numerical and categorical variables.
- **Attention Mechanism:** Identification of important features that influence predictions.
- **Interpretability:** Ability to understand which factors contribute to patient inflow patterns.

The TFT model is trained using historical ETU data and generates baseline predictions of incoming patients for the next shift. Additionally, the model supports demographic segmentation by enabling separate predictions for male and female patient categories.

2.5.2 Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) network is a type of recurrent neural network (RNN) specifically designed to handle sequential data. It is highly effective in capturing short-term dependencies and recent trends in time-series datasets.

In this system, the LSTM model is used to complement the TFT model by focusing on recent fluctuations in patient inflow. While the TFT captures long-term patterns, the LSTM acts as a short-term predictor that reacts quickly to sudden changes in data.

Key features of the LSTM model include:

- **Memory Cells:** Ability to retain important information over multiple time steps.
- **Sequential Learning:** Effective modeling of time-dependent data.
- **Short-Term Pattern Detection:** Identification of recent spikes or drops in patient arrivals.

The LSTM model is trained using a sliding window approach, where a sequence of recent shifts is used as input to predict the next shift's patient inflow. This allows the model to adapt to immediate changes in hospital activity.

2.5.3 Ensemble Prediction Approach

To enhance prediction performance, the outputs of the TFT and LSTM models are combined using an ensemble approach. The ensemble method integrates the strengths of both models to produce a more stable and accurate prediction.

The ensemble process operates as follows:

1. The TFT model generates a prediction based on long-term trends and contextual factors.
2. The LSTM model produces a prediction based on recent historical data.
3. The outputs of both models are combined, typically using averaging or weighted averaging techniques.
4. The final prediction is adjusted to ensure consistency and practicality, such as rounding to the nearest integer.

This combined approach reduces the impact of individual model errors and improves overall reliability. By balancing long-term forecasting with short-term responsiveness, the ensemble model provides a more robust estimate of patient inflow.

The final prediction output serves as the input for the next stage of the system, where bed allocation and triage decision-making are performed based on predicted demand and available capacity.

2.6 Triage and Bed Optimization Logic

The triage and bed optimization logic is the core component of the proposed system, responsible for transforming predicted patient inflow into actionable decisions. While machine learning models provide accurate forecasts of incoming patients, prediction alone is insufficient for real-world hospital operations. Therefore, a practical and reliable mechanism is required to manage capacity constraints and ensure efficient patient flow within the Emergency Treatment Unit (ETU).

To address this requirement, a rule-based triage algorithm was developed to dynamically allocate patients based on available resources. Unlike traditional optimization approaches that rely on complex mathematical solvers, the proposed method uses a deterministic logic structure that ensures reliability, simplicity, and real-time performance.

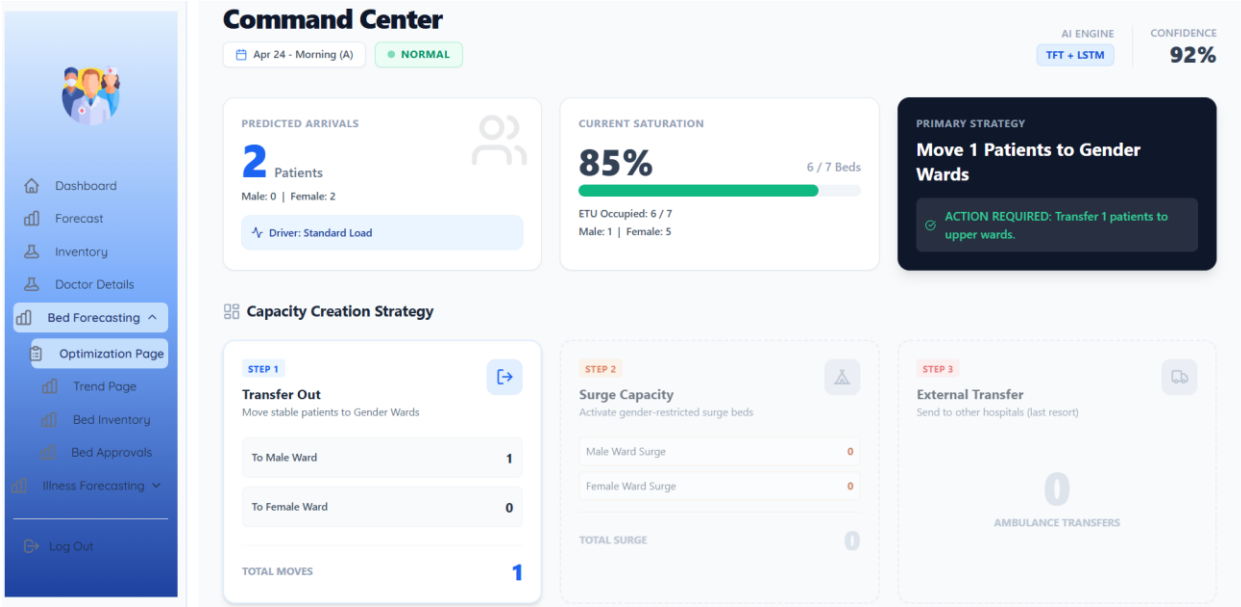

Figure 4- Capacity creation and optimization strategy interface

This interface displays the system-generated triage strategy, including patient transfers to wards, surge bed activation, and external transfers. It provides clear step-by-step actions for managing ETU overflow.

2.6.1 Overflow Detection

The first step in the triage process is to determine whether the ETU can accommodate the predicted number of incoming patients. This is achieved by comparing the current ETU occupancy with the total bed capacity and the predicted patient inflow.

The overflow is calculated using the following logic:

- **Total Expected Patients = Current Occupancy + Predicted Inflow**
- **Overflow = Total Expected Patients − ETU Capacity**

If the overflow value is greater than zero, it indicates that the ETU does not have sufficient capacity to handle incoming patients, and corrective action is required.

2.6.2 Patient Transfer Strategy

Instead of rejecting incoming patients, the system creates capacity by transferring existing patients from the ETU to other available locations. This approach aligns with real hospital workflows, where stabilized patients are moved to wards to free up space for emergency cases.

A key design decision in this system is that **existing ETU patients are transferred**, rather than predicted incoming patients. This ensures that all new patients can be admitted immediately upon arrival.

2.6.3 Gender-Based Routing

In real hospital environments, patient allocation must adhere to gender-based ward structures. Therefore, the system incorporates gender-specific routing to ensure practical feasibility.

- Male patients are transferred only to male wards.
- Female patients are transferred only to female wards.

The system uses current ETU occupancy data (male and female counts) to determine how many patients of each category can be transferred. This ensures that routing decisions are based on actual hospital conditions rather than predictions alone.

2.6.4 Priority-Based Allocation

The triage algorithm follows a prioritized allocation strategy to ensure efficient use of hospital resources. The allocation process is performed in three stages:

1. Standard Ward Allocation

The system first attempts to transfer patients to available beds in standard hospital wards. This is the preferred option, as wards provide the most appropriate environment for continued patient care.

The number of patients transferred is limited by:

- Available ward capacity
- Number of patients in the ETU

2. Surge Bed Allocation

If standard ward capacity is insufficient, the system utilizes surge beds. Surge beds represent temporary or overflow accommodations used during high-demand situations.

This step ensures that additional capacity is created within the hospital before considering external transfers.

3. External Transfer

If both ward and surge capacities are fully utilized, the remaining overflow patients are assigned for transfer to external healthcare facilities.

This is considered the last resort and is only used when all internal resources are exhausted.

2.6.5 Algorithm Behavior Example

To illustrate the operation of the triage algorithm, consider the following scenario:

- ETU Capacity = 7 beds
- Current Occupancy = 6 patients
- Predicted Incoming Patients = 5

Step 1: Overflow Calculation

Total Expected Patients = 6 + 5 = 11

Overflow = 11 − 7 = 4 patients

Step 2: Transfer Execution

The system must transfer 4 existing patients from the ETU.

Assume:

- Male patients in ETU = 3
- Female patients in ETU = 3
- Male ward available beds = 2
- Female ward available beds = 1
- Surge beds available = 2

Step 3: Routing Decisions

- Transfer 2 male patients to male ward
- Transfer 1 female patient to female ward
- Transfer remaining 1 patient to surge bed

This ensures that 4 beds are freed in the ETU, allowing all predicted incoming patients to be accommodated.

2.6.6 Advantages of the Proposed Approach

The proposed triage algorithm offers several advantages:

- **Reliability:** Does not depend on external optimization solvers, avoiding system failures.
- **Real-Time Execution:** Provides immediate decisions without computational delays.
- **Practical Feasibility:** Incorporates real hospital constraints such as gender-based wards.
- **Scalability:** Can be extended to support additional hospital departments and resources.
- **Simplicity:** Easy to implement and maintain compared to complex optimization models.

2.6.7 Integration with Prediction Module

The triage algorithm operates directly on the output of the ensemble prediction model. The predicted patient inflow serves as the input for overflow calculation, which then drives the allocation decisions.

By tightly integrating prediction and triage logic, the system ensures that forecasts are immediately translated into actionable plans. This enables healthcare staff to prepare in advance, reducing overcrowding and improving patient flow within the ETU.

2.7 System Implementation

The proposed temporal ETU hospital bed demand forecasting and optimization system was implemented using a full-stack architecture that integrates a web-based frontend, a backend processing server, and a centralized database. The system is designed to operate in real time, ensuring seamless data flow between components while supporting prediction and triage decision-making processes.

The implementation focuses on scalability, flexibility, and ease of integration with existing hospital systems. Each component of the system is developed using modern technologies that support efficient data handling and processing.

2.7.1 Backend Implementation

The backend of the system is developed using the Python Flask framework. Flask was selected due to its lightweight nature and flexibility in handling RESTful API services. The backend serves as the core processing unit of the system, responsible for executing machine learning models, performing triage calculations, and managing data communication.

The backend performs the following key functions:

- Retrieves real-time data from the database, including ETU occupancy, bed inventory, and surge capacity.
- Processes input data and prepares it for machine learning models.
- Executes the TFT and LSTM models to generate patient inflow predictions.
- Calculates ETU overflow and applies the triage algorithm to determine patient routing.
- Sends processed results to the frontend in JSON format.

The backend is structured in a modular way, separating concerns such as data handling, model execution, and routing logic. This improves maintainability and allows future enhancements without affecting the entire system.

2.7.2 Database Implementation

The system uses MongoDB Atlas (cloud database) as a NoSQL database for storing and managing hospital data. MongoDB was chosen for its flexibility in handling semi-structured data and its ability to scale efficiently.

Several collections are used within the database to manage different types of data:

- **Daily Input Collection:** Stores ETU operational data such as admissions, discharges, and occupancy levels.
- **Prediction History Collection:** Stores historical predictions generated by the system for analysis and evaluation.
- **Bed Inventory Collection:** Maintains information about available beds, including their status (functional or non-functional).

- **Surge Capacity Collection:** Stores data related to temporary surge beds available during peak demand.

MongoDB enables fast data retrieval and supports real-time updates, making it suitable for dynamic hospital environments. The database acts as a central data source that ensures consistency across all system components.

2.7.3 Frontend Implementation

The frontend of the system is developed using React.js, a modern JavaScript library for building user interfaces. The frontend serves as the visual interface through which healthcare staff interact with the system.

The key features of the frontend include:

- Displaying predicted patient inflow for the upcoming shift.
- Showing current ETU occupancy and available bed capacity.
- Presenting recommended actions based on triage decisions.
- Providing visual indicators for system status, such as normal operation or critical overflow conditions.

The interface is designed to be simple and user-friendly, allowing staff to quickly understand system outputs without requiring technical knowledge. React components are used to dynamically update the interface based on data received from the backend.

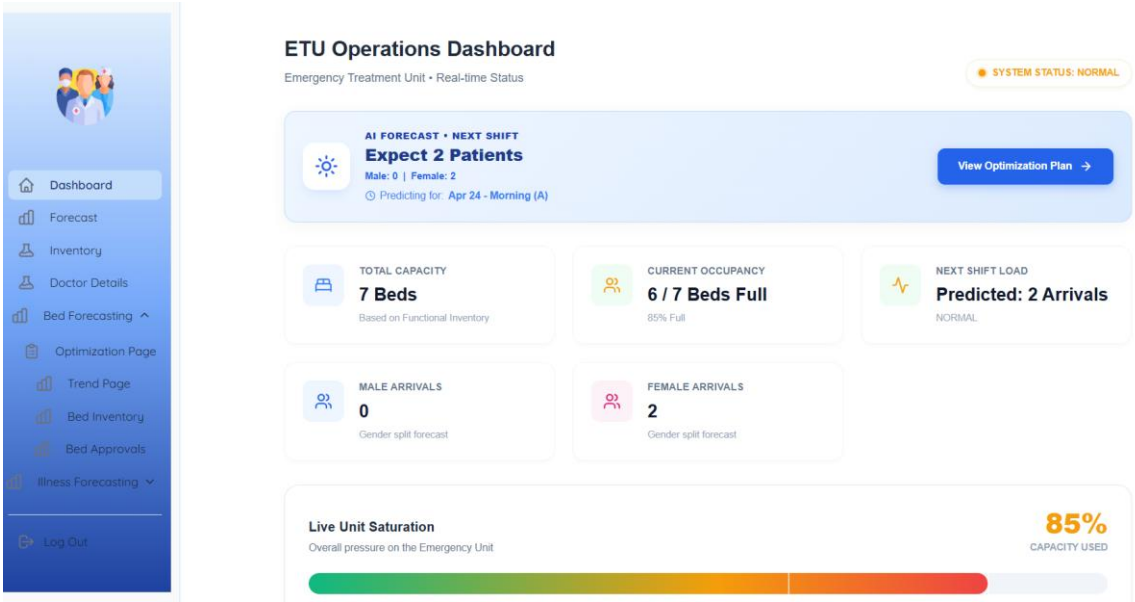

Figure 5- ETU operations dashboard interface

The main dashboard provides real-time visibility of ETU operations, including predicted patient inflow, current occupancy, and system status. It enables healthcare staff to quickly assess capacity and prepare for incoming patients.

2.7.4 API Communication

Communication between the frontend and backend is handled through RESTful APIs. The frontend sends requests to the backend, which processes the data and returns responses in JSON format.

The API workflow operates as follows:

1. The frontend sends a request to the backend to retrieve the latest system data.
2. The backend fetches data from the database and performs necessary computations, including prediction and triage.
3. The processed results are returned to the frontend.
4. The frontend updates the interface to display the latest information.

This request-response mechanism ensures that the system operates in real time and provides up-to-date insights to users.

2.7.5 Integration of System Components

The integration of frontend, backend, and database components ensures a seamless workflow. The frontend continuously interacts with the backend to retrieve updated predictions and triage decisions. The backend, in turn, relies on the database for real-time data and historical records.

This integrated architecture allows the system to function as a unified platform where prediction and decision-making are tightly coupled. As a result, healthcare staff receive accurate and timely information, enabling proactive management of ETU resources.

2.7.6 System Scalability and Deployment

The system is designed with scalability in mind, allowing it to be extended to support additional hospital departments or larger datasets. The use of modular architecture and cloud-compatible technologies enables deployment in both local and cloud environments.

The backend can be deployed on a server environment, while the frontend can be hosted as a web application accessible through standard browsers. MongoDB can be deployed locally or through cloud-based services such as MongoDB Atlas.

This flexibility ensures that the system can be adapted to different hospital infrastructures and requirements.

2.8 System Testing and Evaluation

Testing and implementation are critical phases in ensuring the reliability, accuracy, and practical usability of the proposed ETU hospital bed demand forecasting and optimization system. The system was tested under multiple scenarios to evaluate both its predictive performance and its ability to handle real-world hospital conditions.

2.8.1 Implementation Environment

The system was implemented using a standard development environment with the following technologies:

- **Programming Language:** Python (for backend processing and model execution)
- **Backend Framework:** Flask
- **Frontend Framework:** React.js
- **Database:** MongoDB
- **Machine Learning Libraries:** PyTorch, Pandas, NumPy

The backend server was deployed locally during development and testing, while the frontend application interacted with the backend through RESTful API calls. The database was configured to store both historical and real-time operational data.

2.8.2 Functional Testing

Functional testing was conducted to verify that each component of the system operates correctly according to its intended functionality. The following aspects were tested:

- Correct retrieval of data from the database
- Accurate execution of machine learning models
- Proper calculation of ETU overflow
- Correct routing of patients based on triage logic
- Successful communication between frontend and backend

Each module was tested individually and then integrated to ensure that the complete system performs as expected.

2.8.3 Scenario-Based Testing

To evaluate the practical effectiveness of the system, several real-world scenarios were simulated. These scenarios included:

Normal Operation Scenario

- Predicted patient inflow is within ETU capacity
- System correctly identifies that no transfers are required

Moderate Overflow Scenario

- Predicted inflow exceeds available beds by a small margin
- System transfers patients to available ward beds

High Overflow Scenario

- Predicted inflow significantly exceeds capacity
- System utilizes both ward beds and surge beds

Critical Overflow Scenario

- All internal resources (ward and surge beds) are exhausted
- System recommends external patient transfers

These scenarios demonstrate the system's ability to adapt to different levels of demand and ensure continuity of patient care.

2.8.4 Edge Case Testing

The system was also tested under extreme and edge-case conditions to evaluate its robustness. These included:

- **Zero Available Beds:** System correctly routes all overflow patients to surge or external transfers
- **Incomplete Data:** Missing values are handled using preprocessing techniques without system failure
- **High Patient Surge:** System continues to function without delay even under extreme inflow conditions

- **Imbalanced Gender Distribution:** System correctly applies gender-based routing constraints

These tests confirm that the system can operate reliably even under unpredictable conditions.

2.8.5 Performance Evaluation

The performance of the system was evaluated based on two key factors:

Prediction Accuracy

The ensemble model (TFT + LSTM) demonstrated improved prediction stability compared to individual models. By combining long-term trends with short-term patterns, the system produced reliable forecasts for incoming patient demand.

Triage Execution Efficiency

The triage algorithm executed routing decisions in real time with minimal computational overhead. Unlike traditional optimization models that may experience delays, the rule-based approach ensured immediate response without system failures.

2.8.6 System Reliability

One of the primary goals of this research was to develop a reliable system that does not fail under high-pressure conditions. Testing results indicate that the system consistently produces valid outputs without crashing or generating infeasible solutions.

The deterministic nature of the triage algorithm ensures that a valid routing plan is always generated, regardless of the input conditions. This is particularly important in emergency healthcare environments where system reliability is critical.

2.8.7 Summary of Testing

Overall, the testing phase confirms that the proposed system is:

- Accurate in predicting patient inflow
- Reliable in handling capacity constraints
- Robust under extreme and edge-case conditions
- Practical for real-world hospital applications

The system successfully meets its intended objectives and demonstrates strong potential for improving ETU management and decision-making processes.

3. RESULTS AND DISCUSSION

3.1 Results

The implementation of the temporal ETU hospital bed demand forecasting and optimization system produced significant results in both predictive accuracy and operational decision-making. The system was evaluated using historical data and simulated real-world scenarios to assess its effectiveness in managing Emergency Treatment Unit (ETU) capacity. This demonstrates that the proposed system not only predicts patient demand but also generates actionable operational strategies, effectively bridging the gap between forecasting and real-time decision-making.

The results are presented in terms of prediction performance, system outputs, and triage decision execution. These outcomes demonstrate how the system transforms predicted patient demand into actionable strategies for hospital resource management.

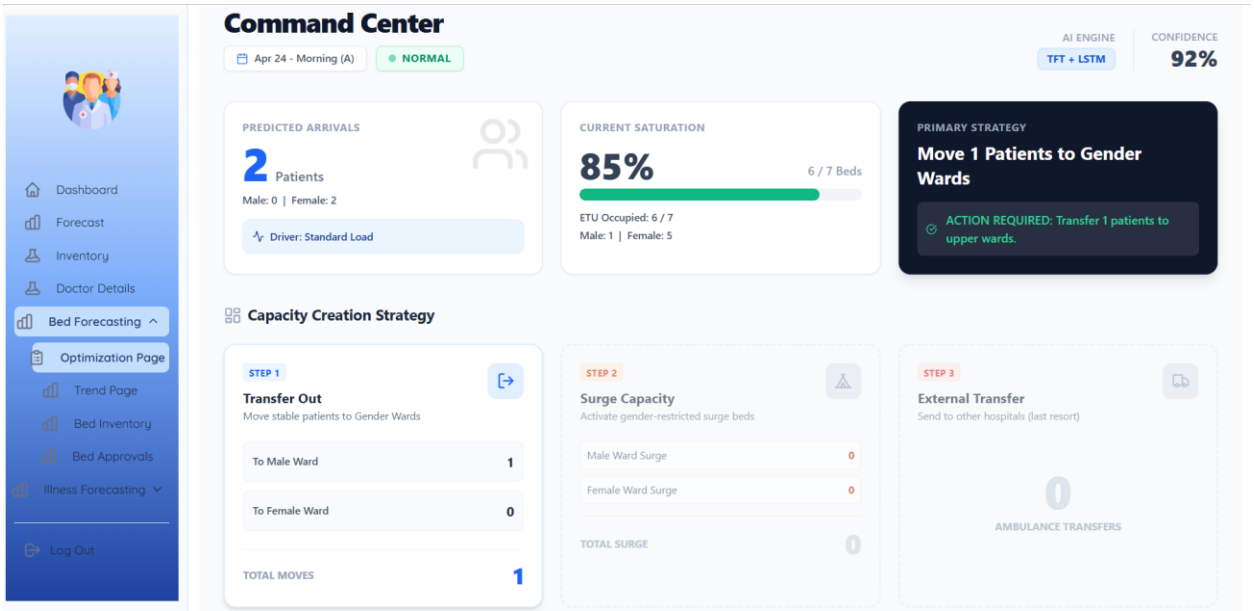

Figure 6- System-generated bed optimization and triage plan

The optimization interface presents the system-generated triage plan based on predicted patient inflow and current ETU capacity. It clearly outlines the required actions, including transferring patients to gender-specific wards, activating surge capacity if necessary, and minimizing external transfers. This demonstrates how the system converts predictions into actionable decisions in real time.

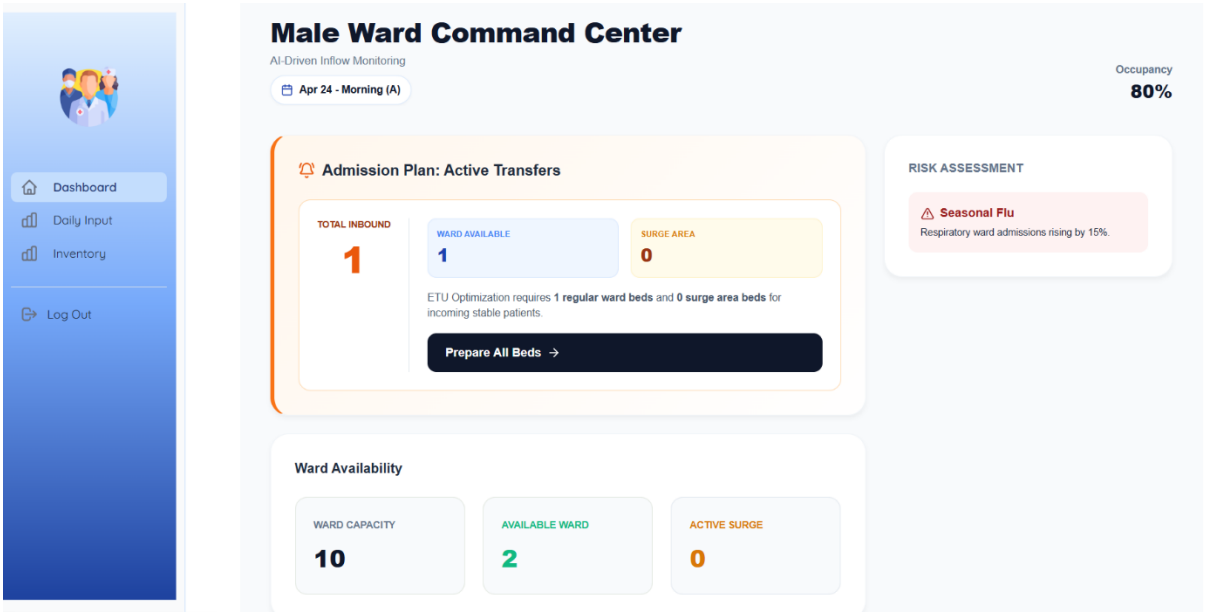

Figure 7-Male ward command center view

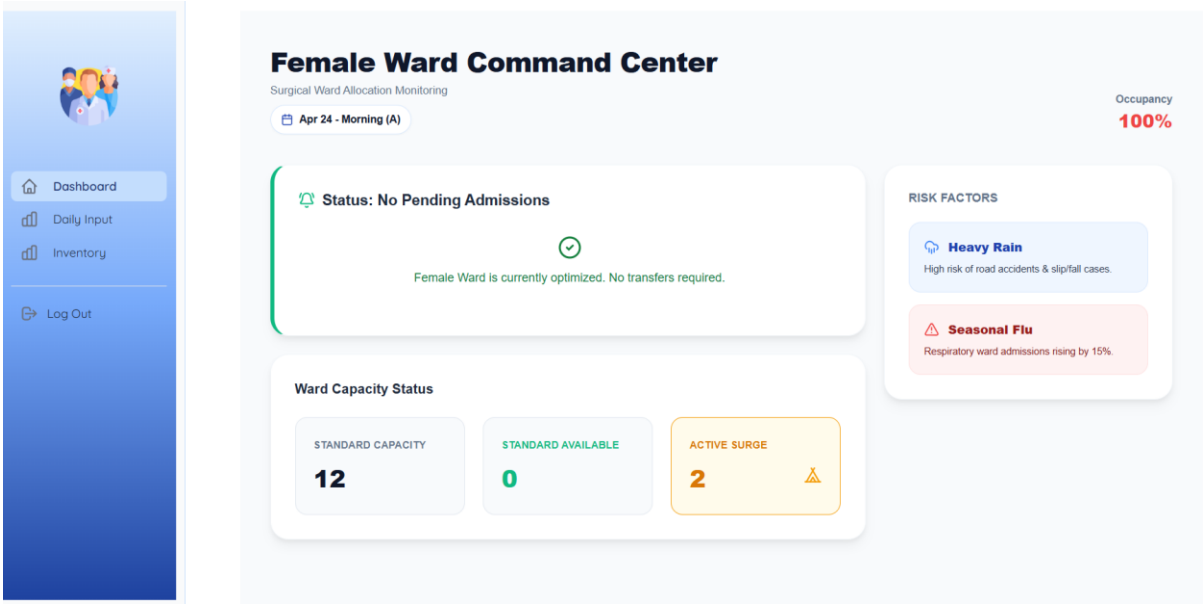

Figure 8- Female ward command center view

3.1.1 Prediction Results

The ensemble machine learning model, which combines the Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM), was used to predict incoming patient demand for upcoming shifts. The model successfully captured both long-term trends and short-term variations in patient inflow.

The TFT model demonstrated strong performance in identifying patterns related to time, seasonal variations, and contextual factors such as weather and shift timing. Meanwhile, the LSTM model effectively captured recent fluctuations in patient arrivals, allowing the system to respond to sudden changes.

The ensemble approach produced more stable and reliable predictions compared to using either model independently. By combining the outputs of both models, the system reduced prediction variance and improved overall consistency. The final predicted values were rounded to practical integers, making them suitable for real-world decision-making.

3.1.2 System Output Results

The system generates multiple outputs that provide a comprehensive view of hospital operations. These outputs are displayed through the frontend dashboard and include:

- **Predicted Patient Inflow:** Number of patients expected to arrive in the next shift
- **Current ETU Occupancy:** Number of beds currently occupied
- **Available Capacity:** Remaining bed capacity in the ETU
- **Overflow Calculation:** Number of patients exceeding available capacity
- **Triage Recommendations:** Number of patients to be transferred and their destinations

These outputs provide healthcare staff with a clear and actionable plan, enabling them to prepare for incoming patients in advance.

3.1.3 Triage Execution Results

The triage algorithm was tested under various simulated scenarios to evaluate its effectiveness in managing overflow situations. The results show that the system consistently produces valid and practical routing decisions.

In scenarios where patient inflow did not exceed capacity, the system correctly identified that no action was required. In overflow scenarios, the system successfully calculated the number of patients to be transferred and allocated them according to available resources.

The algorithm effectively prioritized standard ward allocation, followed by surge beds, and finally external transfers when necessary. This prioritized approach ensures optimal utilization of hospital resources.

3.1.4 Example Scenario Results

To demonstrate the system's behavior, consider the following example:

- ETU Capacity = 7 beds
- Current Occupancy = 5 patients
- Predicted Incoming Patients = 6

Step 1: Overflow Calculation

Total Expected Patients = 5 + 6 = 11

Overflow = 11 − 7 = 4 patients

Step 2: Triage Decision

- Transfer 2 patients to available ward beds
- Transfer 1 patient to surge bed
- Transfer 1 patient to external facility

This result shows how the system dynamically adjusts its decisions based on available resources, ensuring that all incoming patients can be accommodated.

3.1.5 System Performance Summary

The results indicate that the system performs effectively across different operational conditions. Key observations include:

- The ensemble model provides accurate and stable predictions
- The triage algorithm consistently produces valid routing decisions
- The system operates in real time without computational delays
- The outputs are practical and easy for healthcare staff to interpret

Overall, the system successfully demonstrates its ability to integrate prediction and decision-making into a unified framework, providing a proactive solution for ETU management.

3.2 Research Findings

The development and evaluation of the temporal ETU hospital bed demand forecasting and optimization system revealed several important findings related to hospital resource management, predictive analytics, and automated decision-making. These findings highlight both the effectiveness of the proposed approach and the practical challenges associated with traditional hospital operations.

3.2.1 Effectiveness of Ensemble Prediction

One of the key findings of this research is the effectiveness of combining multiple machine learning models for time-series forecasting. The ensemble approach, which integrates the Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM) models, produced more stable and reliable predictions compared to using a single model.

The TFT model was highly effective in capturing long-term patterns and contextual influences, such as weather conditions and shift variations. On the other hand, the LSTM model demonstrated strong performance in detecting short-term fluctuations in patient inflow. By combining these models, the system was able to balance both long-term trends and immediate changes, resulting in improved prediction accuracy.

This finding confirms that ensemble learning is a suitable approach for handling complex and dynamic datasets in healthcare environments.

3.2.2 Importance of Integrating Prediction with Decision-Making

Another significant finding is that prediction alone is not sufficient for practical hospital management. While machine learning models can accurately estimate patient inflow, hospitals require actionable decisions to manage capacity constraints effectively.

The integration of prediction with the triage and bed optimization algorithm proved to be essential. By linking predicted patient demand directly with routing decisions, the system was able to provide a complete solution rather than isolated analytical outputs.

This demonstrates that effective healthcare systems must combine predictive analytics with operational decision-making to achieve meaningful impact.

3.2.3 Limitations of Manual Data Management

The research also highlighted the limitations of traditional manual data entry processes in hospitals. During simulated scenarios, it was observed that missing or delayed updates in bed occupancy data can lead to incorrect assumptions about hospital capacity.

Manual data entry introduces risks such as human error, data inconsistency, and delays in updating critical information. These issues can significantly reduce the effectiveness of decision-support systems that rely on accurate and timely data.

This finding emphasizes the need for systems that minimize dependency on manual inputs and instead utilize automated or data-driven approaches.

3.2.4 Practical Value of Rule-Based Triage Logic

The implementation of a rule-based triage algorithm proved to be highly effective in handling real-time decision-making. Unlike complex optimization techniques that may fail under extreme conditions, the deterministic logic used in this system consistently produced valid and practical results.

The algorithm's ability to handle overflow scenarios, apply gender-based routing, and prioritize resource allocation ensures that it aligns closely with real hospital workflows. This finding demonstrates that simple, well-designed rule-based systems can be more reliable and practical than complex optimization models in dynamic environments.

3.2.5 Impact of Real-World Constraints

Incorporating real-world constraints, such as gender-based ward allocation and limited surge capacity, significantly improved the practicality of the system. Many existing models do not consider these constraints, resulting in solutions that are not applicable in real hospital settings.

By including these constraints, the system produced decisions that are both realistic and implementable. This highlights the importance of designing systems that reflect actual operational conditions rather than purely theoretical models.

3.2.6 Proactive vs Reactive Hospital Management

A major finding of this research is the advantage of proactive management over reactive approaches. Traditional systems respond to overcrowding after it occurs, whereas the proposed system anticipates future demand and prepares resources in advance.

By predicting patient inflow and generating preemptive action plans, the system reduces the likelihood of overcrowding and improves overall efficiency. This shift from reactive to proactive decision-making represents a significant improvement in hospital management strategies.

3.2.7 Summary of Findings

Overall, the key findings of this research can be summarized as follows:

- Ensemble machine learning improves prediction accuracy and stability
- Prediction must be integrated with decision-making for practical use
- Manual data entry is a major limitation in traditional systems
- Rule-based triage logic provides reliable and real-time solutions
- Incorporating real-world constraints enhances system practicality
- Proactive systems significantly improve hospital resource management

These findings confirm the effectiveness of the proposed system and its potential to improve ETU operations in real-world healthcare environments.

3.3 Discussion

The results and findings of this research demonstrate the effectiveness of integrating machine learning with practical decision-making mechanisms in the context of Emergency Treatment Unit (ETU) management. This section discusses the implications of the results, evaluates the strengths of the proposed system, and highlights its limitations in real-world applications.

3.3.1 Effectiveness of the Proposed System

The proposed system successfully addresses the limitations identified in traditional hospital management approaches. By combining predictive analytics with automated triage logic, the system provides a complete solution for managing patient inflow and bed allocation.

The ensemble machine learning model proved effective in capturing both long-term and short-term patterns in patient arrivals. This allowed the system to generate reliable predictions that reflect real-world conditions. More importantly, these predictions were directly translated into actionable decisions through the triage algorithm, ensuring that the system is not limited to theoretical analysis but can be applied in practical scenarios.

The ability of the system to operate in real time and provide immediate recommendations enhances its usability in high-pressure environments such as ETUs.

3.3.2 Comparison with Traditional Approaches

Traditional hospital management systems rely heavily on manual data entry and reactive decision-making. In such systems, actions are taken only after overcrowding occurs, which can lead to delays in treatment and increased stress on healthcare staff.

In contrast, the proposed system adopts a proactive approach. By predicting patient inflow in advance and preparing a corresponding action plan, the system allows healthcare staff to respond before critical situations arise. This proactive strategy improves efficiency and reduces the risk of overcrowding.

Furthermore, unlike conventional optimization models that depend on complex mathematical solvers, the rule-based triage algorithm used in this system provides consistent and reliable results. This makes it more suitable for real-time applications where speed and reliability are critical.

3.3.3 Practical Implications

The system has significant practical implications for hospital operations. By reducing reliance on manual data entry, the system minimizes human error and ensures more accurate representation of

hospital capacity. This is particularly beneficial in emergency situations where timely and accurate information is essential.

The incorporation of real-world constraints, such as gender-based ward allocation and surge bed usage, ensures that the system's recommendations can be directly implemented without additional adjustments. This increases the practicality and usability of the system in real healthcare environments.

Additionally, the user-friendly interface enables healthcare staff to easily interpret system outputs, facilitating quick and informed decision-making.

3.3.4 Strengths of the System

The proposed system has several key strengths:

- **Integrated Approach:** Combines prediction and decision-making into a single framework
- **Real-Time Operation:** Provides immediate outputs without delays
- **Reliability:** Uses a deterministic triage algorithm that avoids system failures
- **Practical Design:** Incorporates real-world hospital constraints
- **Scalability:** Can be extended to other hospital departments

These strengths contribute to the overall effectiveness and applicability of the system.

3.3.5 Limitations of the System

Despite its advantages, the system has certain limitations that should be considered:

- **Dependence on Data Quality:** The accuracy of predictions depends on the quality and completeness of the dataset
- **Simplified Assumptions:** Some aspects, such as patient discharge rates, may be estimated rather than measured in real time
- **Limited Scope:** The system is primarily designed for ETU operations and may require modifications to support other hospital units

- **Lack of Real-Time Sensor Data:** The system currently relies on recorded data rather than live tracking technologies

These limitations indicate areas where further improvements can be made.

3.3.6 Overall Evaluation

Overall, the proposed system demonstrates a significant improvement over traditional ETU management approaches. By integrating machine learning with a practical triage algorithm, the system provides a reliable and efficient solution for managing patient inflow and bed allocation.

The results confirm that a proactive, data-driven approach can enhance hospital operations, reduce overcrowding, and improve patient care. While there are limitations, the system provides a strong foundation for future development and real-world implementation.

3.4 Summary of Contributions

This research presents a comprehensive solution for managing Emergency Treatment Unit (ETU) operations through the integration of machine learning and automated decision-making. The main contribution of this study is the development of a unified system that not only predicts patient inflow but also provides practical strategies for bed allocation and patient routing.

The contributions of this research can be summarized as follows:

3.4.1 Development of an Ensemble Prediction Model

A significant contribution of this research is the implementation of an ensemble machine learning model that combines the strengths of Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM) networks. This hybrid approach enables the system to capture both long-term temporal patterns and short-term fluctuations in patient inflow.

The ensemble model improves prediction stability and accuracy, making it more suitable for real-world hospital environments where demand patterns are highly dynamic and uncertain.

3.4.2 Integration of Prediction with Decision-Making

Unlike traditional systems that treat prediction and optimization as separate processes, this research integrates both components into a single framework. The predicted patient inflow is directly used to drive triage and bed allocation decisions, ensuring that forecasts are translated into actionable outcomes.

This integration represents a key advancement in transforming predictive analytics into practical decision-support systems.

3.4.3 Design of a Real-Time Triage and Routing Algorithm

The research introduces a rule-based triage algorithm that dynamically allocates patients based on available resources. The algorithm prioritizes standard ward beds, followed by surge beds, and finally external transfers when necessary.

The use of a deterministic approach ensures reliability and real-time performance, making the system suitable for high-pressure emergency environments.

3.4.4 Incorporation of Real-World Constraints

Another important contribution is the inclusion of practical constraints within the system design. These include gender-based ward allocation, limited bed capacity, and surge bed availability.

By incorporating these constraints, the system produces solutions that are realistic and directly applicable in hospital settings, bridging the gap between theoretical models and real-world implementation.

3.4.5 Development of a Full-Stack System

The research also demonstrates the implementation of a complete system using modern technologies. The backend is developed using Python Flask, the frontend is built with React.js, and MongoDB is used for data storage.

This full-stack implementation showcases the practical feasibility of the proposed solution and provides a foundation for real-world deployment.

3.4.6 Contribution to Proactive Healthcare Management

Finally, the system contributes to shifting hospital management from a reactive approach to a proactive one. By predicting patient inflow and preparing resources in advance, the system reduces overcrowding and improves overall efficiency.

This contribution highlights the potential of integrating AI-driven solutions into healthcare systems to enhance decision-making and patient care.

4. CONCLUSIONS AND RECOMMENDATIONS

4.1 Conclusion

This research focused on addressing the challenges associated with patient inflow management and bed allocation in Emergency Treatment Units (ETUs). Hospitals often face difficulties in handling unpredictable patient demand, leading to overcrowding, inefficient resource utilization, and increased pressure on healthcare staff. Traditional methods, which rely heavily on manual data entry and reactive decision-making, are insufficient to handle the dynamic nature of emergency healthcare environments.

To overcome these challenges, this study proposed a temporal ETU hospital bed demand forecasting and optimization system that integrates machine learning with automated decision-making. The system utilizes an ensemble approach combining Temporal Fusion Transformer (TFT) and Long Short-Term Memory (LSTM) models to predict patient inflow for upcoming shifts. These predictions are then used as inputs for a rule-based triage algorithm that dynamically allocates patients based on available hospital resources.

The results of the study demonstrate that the proposed system is capable of accurately predicting patient demand and effectively managing capacity constraints. By transferring existing patients to wards, surge beds, or external facilities, the system ensures that sufficient space is available for

incoming patients. This proactive approach significantly reduces the risk of overcrowding and improves overall operational efficiency.

One of the key strengths of this research is the integration of prediction and decision-making into a unified framework. Unlike traditional systems that treat these components separately, the proposed system provides end-to-end functionality, from forecasting to actionable recommendations. Additionally, the inclusion of real-world constraints, such as gender-based ward allocation and limited surge capacity, enhances the practicality and applicability of the system.

The implementation of the system using modern technologies, including Flask, React.js, and MongoDB, further demonstrates its feasibility for real-world deployment. The system is designed to operate in real time, providing healthcare staff with timely and reliable information to support decision-making.

In conclusion, this research successfully demonstrates that combining machine learning with practical triage logic can significantly improve ETU management. The proposed system provides a reliable, efficient, and scalable solution that has the potential to enhance patient care and hospital operations. It represents a step towards the adoption of intelligent, data-driven systems in modern healthcare environments.

4.2 Future Work and Recommendations

Although the proposed temporal ETU hospital bed demand forecasting and optimization system demonstrates strong performance and practical applicability, there are several areas where further improvements and extensions can be made. These enhancements can increase the system's accuracy, scalability, and real-world impact in healthcare environments.

4.2.1 Integration with Real-Time Data Sources

One of the key areas for future improvement is the integration of real-time data sources. Currently, the system relies on recorded operational data, which may involve some level of delay. Incorporating real-time data collection methods, such as IoT-based bed monitoring systems or automated hospital information systems, can significantly improve the accuracy of capacity estimation and decision-making.

Real-time integration would allow the system to continuously update bed availability, patient status, and resource utilization, enabling more dynamic and responsive operations.

4.2.2 Expansion to Multiple Hospital Units

The current system is primarily designed for Emergency Treatment Units (ETUs). Future work can extend the system to cover additional hospital departments, such as Intensive Care Units (ICUs), general wards, and outpatient departments.

Expanding the system to multiple units would allow hospitals to manage resources across the entire organization, improving overall efficiency and coordination between departments.

4.2.3 Advanced Optimization Techniques

While the current rule-based triage algorithm provides reliable and real-time decision-making, future research can explore the integration of advanced optimization techniques, such as Mixed-Integer Linear Programming (MILP) or reinforcement learning.

These approaches may provide more optimized solutions under certain conditions, especially when dealing with large-scale hospital systems. However, they must be carefully designed to ensure stability and real-time performance.

4.2.4 Incorporation of Additional Data Features

Future improvements can include the incorporation of additional data sources to enhance prediction accuracy. These may include:

- Patient medical history
- Severity levels of cases
- Staff availability and workload
- Seasonal disease trends

Including such features would allow the system to make more informed and context-aware decisions.

4.2.5 Explainability and Visualization Enhancements

Another important area for future work is improving the explainability of model predictions. Providing clear explanations for why certain predictions or decisions are made can increase trust and acceptance among healthcare professionals.

Additionally, enhancing the visualization capabilities of the system, such as interactive dashboards and trend analysis charts, can improve user experience and facilitate better decision-making.

4.2.6 Deployment in Real Hospital Environments

The ultimate goal of this research is real-world implementation. Future work should focus on deploying the system in an actual hospital environment to evaluate its performance under real operational conditions.

This would involve collaboration with healthcare institutions, integration with existing hospital systems, and validation using live data. Real-world deployment would provide valuable insights into system performance and potential improvements.

4.2.7 Continuous Model Improvement

Machine learning models require continuous updates to maintain their accuracy over time. Future work can include implementing automated model retraining mechanisms that update the models as new data becomes available.

This ensures that the system adapts to changing patterns in patient inflow and maintains high prediction accuracy.

4.2.8 Summary of Recommendations

In summary, future improvements should focus on:

- Integrating real-time data sources
- Expanding the system to multiple hospital units
- Exploring advanced optimization techniques
- Incorporating additional data features

- Enhancing explainability and visualization
- Deploying the system in real-world environments
- Implementing continuous model updates

These enhancements will further strengthen the system and increase its potential impact in modern healthcare systems.

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